

Complete graph K_5

Every pair of distinct vertices is joined by an edge

What the picture shows

Degree of every vertex

$$\deg(v) = 5 - 1 = 4$$

Number of edges

$$|\mathcal{E}| = \frac{5(4)}{2} = 10$$

Fraction of possible edges

$$\rho(K_5) = 1$$

 edges incident to vertex 1

